

14C CHEMICAL KINETICS



UNIT 02: HOW REACTIONS HAPPEN

CHEMISTRY 1310

LEARNING OBJECTIVES

CHAPTER 14

1. Express reactions rates in relation to reaction stoichiometry.
2. Use experimental data to write rate laws for chemical reactions.
3. Predict the concentration and time values of reactions using first- and second-order integrated rate laws.
4. Graph data to identify the order of a reaction.
5. Define and apply the terms half-life, first-order reaction, and second-order reaction.
6. Explain collision theory using appropriate terminology.
7. Apply the Arrhenius equation to calculate activation energy and/or the value of the rate constant at various temperatures.
8. Define and discuss reaction mechanisms.
9. Write rate laws for elementary steps.
10. Identify intermediates in a reaction mechanism.
11. Discuss how catalysts work to speed up chemical reactions

Integrated Rate Laws and Half-Lives

14.3

Integrated rate laws are equations that relate concentrations and time (as opposed to concentration and rate for *differential* rate laws).

- Derived by taking the integral of the *differential* rate law.
- Since the differential rate laws for zeroth, first, and second orders are all different, we also end up with different versions of the integrated rate laws.

Three important results of the integrated rate laws:

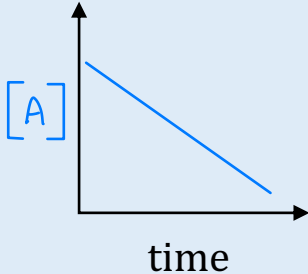
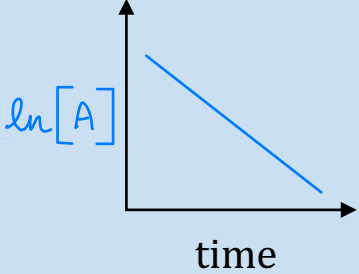
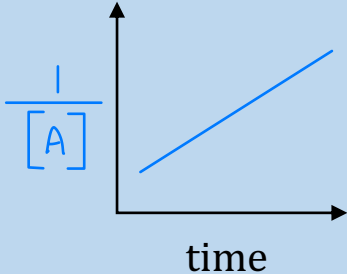
- 1) Predict concentrations at different times.
- 2) Determine half-life (time, $t_{1/2}$).
- 3) Identify orders by plotting concentration and time data.

Integrated Rate Laws and Half-Lives

14.3

For a generic reaction with a single reactant A : $aA \rightarrow \text{products}$

$$\text{Rate} = k[A]^m$$

	Zeroth Order ($m = 0$)	First Order ($m = 1$)	Second Order ($m = 2$)
Rate Law	$\text{Rate} = k[A]^0 = k$	$\text{Rate} = k[A]^1$	$\text{Rate} = k[A]^2$
Units of k	$M \cdot s^{-1}$	s^{-1}	$M^{-1} \cdot s^{-1}$
Integrated Rate Law	$[A]_t = -kt + [A]_0$	$\ln[A]_t = -kt + \ln[A]_0$	$\frac{1}{[A]_t} = kt + \frac{1}{[A]_0}$
Linear Plot	slope = $-k$ 	slope = $-k$ 	slope = k 
Half-Life	$t_{1/2} = \frac{[A]_0}{2k}$	$t_{1/2} = \frac{\ln 2}{k} \cong \frac{0.693}{k}$	$t_{1/2} = \frac{1}{k[A]_0}$

Integrated Rate Laws and Half-Lives

14.3

A **half-life** ($t_{1/2}$) is the time it takes for concentration to decrease to *half* its original value.

- In other words, the half-life is the time required for $[A]$ to become equal to $\frac{1}{2}[A]_0$.
- This can be used with the different integrated rate laws to determine a half-life expression for each order of reaction.

For a **first-order** reaction:

1) *Integrated rate law*

$$\ln[A]_t = -kt + \ln[A]_0$$

"initial"

2) *Rearrange*

$$kt = \ln \frac{[A]_0}{[A]_t}$$

3) *Substitute*

at $t_{1/2}$: $[A]_{t_{1/2}} = \frac{1}{2}[A]_0$

$$kt_{1/2} = \ln \frac{[A]_0}{\frac{1}{2}[A]_0}$$

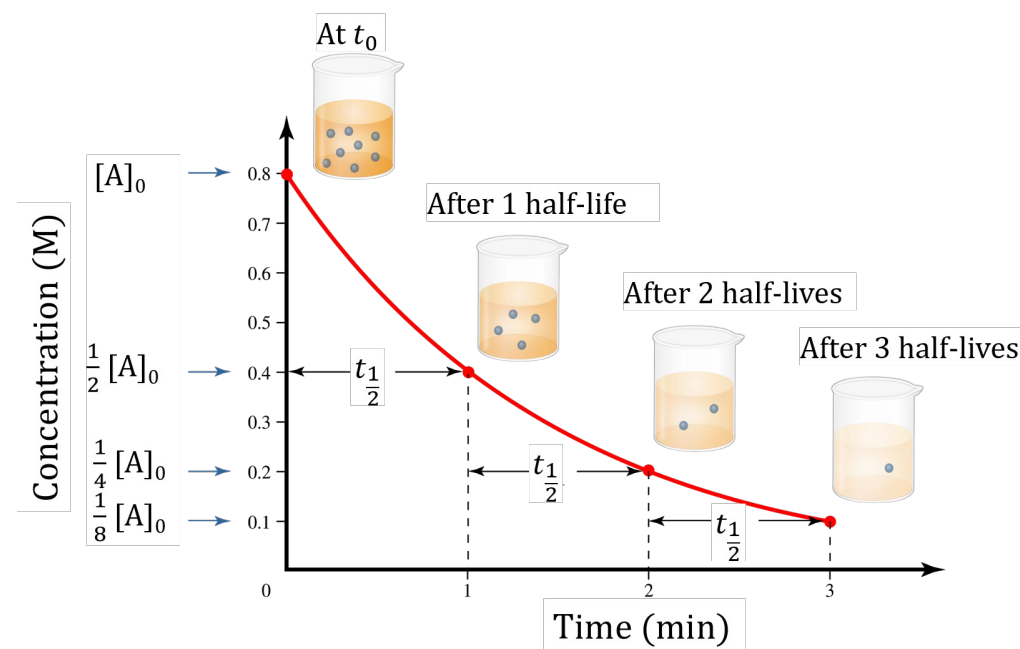
$$kt_{1/2} = \ln \frac{1}{\frac{1}{2}}$$

$$kt_{1/2} = \ln 2$$

4) *Simplify*

$$t_{1/2} = \frac{\ln 2}{k}$$

FIGURE 14.11 Half-Life for a First-Order Reaction



EXAMPLE PROBLEM

Learning Objective(s): 14.3

At a given temperature, N_2O_5 undergoes first-order decomposition with $k = 0.0052 \text{ s}^{-1}$.



If the initial concentration was 0.040 M, what is the concentration (in M) of N_2O_5 after 10.0 minutes?


$$10.0 \text{ min} \times \frac{60 \text{ s}}{1 \text{ min}} = 600. \text{ s}$$

integrated 1st order rate law: $\ln[\text{N}_2\text{O}_5]_t = -kt + \ln[\text{N}_2\text{O}_5]_0$

$$\ln[\text{N}_2\text{O}_5]_{600 \text{ s}} = - (0.0052 \text{ s}^{-1})(600 \text{ s}) + \ln(0.040)$$

$$\ln[\text{N}_2\text{O}_5]_{600 \text{ s}} = -6.334$$

$$[\text{N}_2\text{O}_5]_{600 \text{ s}} = \exp(-6.334)$$

$$[\text{N}_2\text{O}_5]_{600 \text{ s}} = 0.0018 \text{ M}$$

IN-CLASS QUESTION 1

Learning Objective(s): 14.3, 14.5

At 550 K, the decomposition of a $[NO_2]_0 = 0.250 \text{ M}$ sample of NO_2 into nitrogen and oxygen gases has $k = 0.312 \text{ M}^{-1} \cdot \text{hr}^{-1}$.

How long (in hours) would it take for $[NO_2]_t = 18.2\%$ of the NO_2 to decompose?

At beginning, 100% NO_2 , so $[NO_2]_0 = 0.250 \text{ M}$

After time, 18.2% NO_2 gone, so $[NO_2]_t = 0.250 \text{ M} - (0.182)(0.250 \text{ M})$

$[NO_2]_t = 0.2045 \text{ M}$ remains

Based on units of $k \rightarrow 2^{\text{nd}}$ order

integrated 2^{nd} order rate law:

$$\frac{1}{[NO_2]_t} = kt + \frac{1}{[NO_2]_0}$$
$$\frac{1}{0.2045 \text{ M}} = (0.312 \text{ M}^{-1} \cdot \text{hr}^{-1})t + \frac{1}{0.250 \text{ M}}$$
$$0.889_9 \text{ M}^{-1} = (0.312 \text{ M}^{-1} \cdot \text{hr}^{-1})t$$

$$t = 2.85 \text{ hr}$$

IN-CLASS QUESTION 2

Learning Objective(s): 14.3, 14.5

The half-life for the following first-order reaction is $t_{1/2}$ 688 hours at 1000 °C.



Calculate the rate constant (k) at 1000°C. 1st order half-life $\longrightarrow t_{1/2} = \frac{\ln(2)}{k}$

$$688 \text{ hr} = \frac{\ln(2)}{k}$$

$$k = 0.001007 \text{ hr}^{-1}$$

Then, calculate the mass (in grams) of CS₂ that remains after 48 hours if the initial mass was 3.0 grams.

integrated 1st order rate law: $\ln[\text{CS}_2]_t = -kt + \ln[\text{CS}_2]_0$

$$\ln[\text{CS}_2]_{48 \text{ hr}} = -(0.001007 \text{ hr}^{-1})(48 \text{ hr}) + \ln(3.0)$$

$$\ln[\text{CS}_2]_{48 \text{ hr}} = 1.05$$

$$[\text{CS}_2]_{48 \text{ hr}} = 2.9 \text{ g}$$

Note A different way to express first-order integrated rate law.

$$kt = \ln \frac{[A]_0}{[A]_t}$$

Just a ratio, so if units match, then
[A] can be masses, moles, pressures, etc.

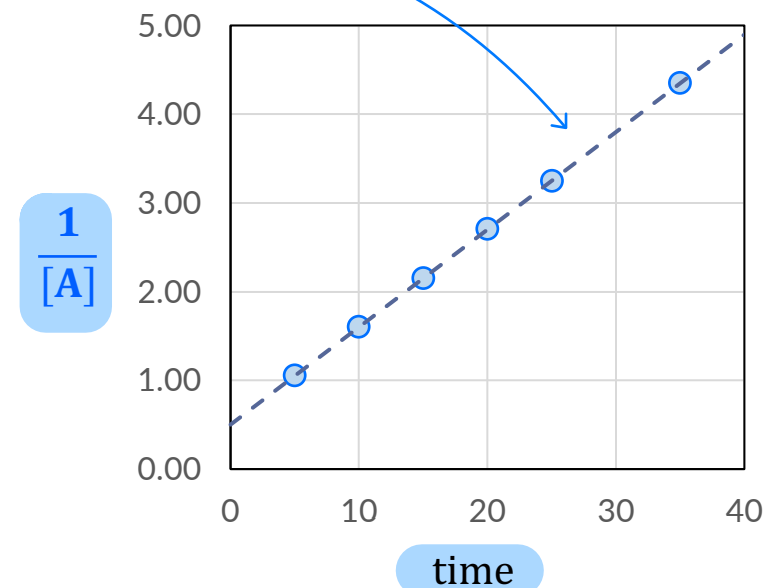
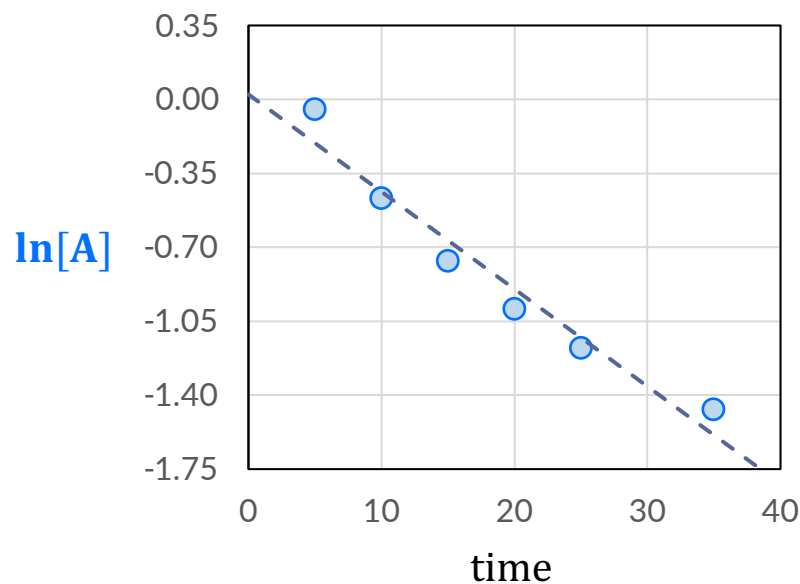
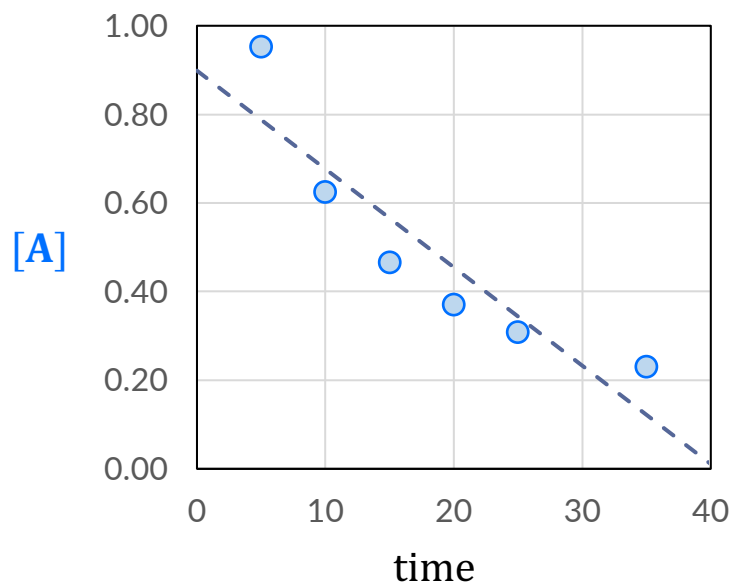
Integrated Rate Laws and Half-Lives

14.3

How to graphically determine orders from plots of concentration-time data?

Plot the data *three* ways, and the **most linear** plot determines the reaction order.

- If a plot of $[A]$ vs. time is linear, the reaction is *zeroth order*.
- If a plot of $\ln[A]$ vs. time is linear, the reaction is *first order*.
- If a plot of $1/[A]$ vs. time is linear, the reaction is *second order*.



IN-CLASS QUESTION 3

Learning Objective(s): 14.3, 14.4, 14.5

Given the plot, calculate the time (in seconds) required for the concentration of A to decrease from 1.00 M to 0.45 M.

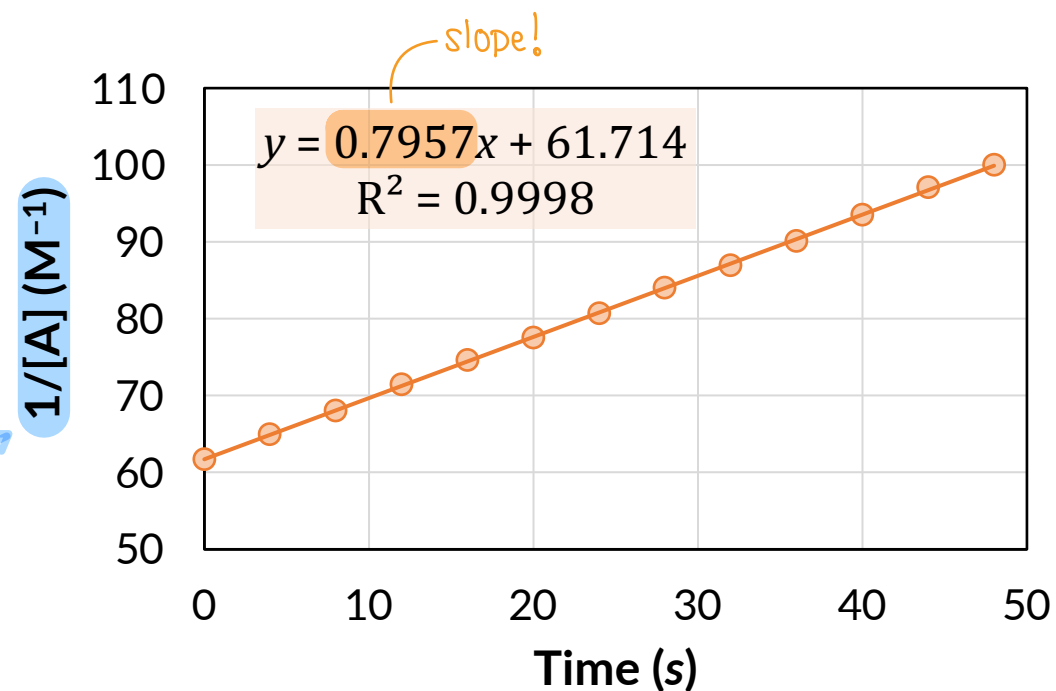


Based on linear plot of $\frac{1}{[A]}$ vs. time $\rightarrow$ 2nd order
 $\rightarrow$ slope = $k = 0.7957 \text{ M}^{-1} \cdot \text{s}^{-1}$

integrated 2nd order rate law:

$$\frac{1}{[A]_t} = kt + \frac{1}{[A]_0}$$
$$\frac{1}{0.45 \text{ M}} = (0.7957 \text{ M}^{-1} \cdot \text{s}^{-1})t + \frac{1}{1.00 \text{ M}}$$
$$1.2_2 \text{ M}^{-1} = (0.7957 \text{ M}^{-1} \cdot \text{s}^{-1})t$$

$$t = 1.5 \text{ s}$$



BONUS QUESTION

Assessed on: Fall 2021, Exam 3

The first-order reaction $A \rightarrow B$ has a $k = 0.047 \text{ s}^{-1}$. If the initial concentration of A is 4.99 M, what is the concentration (in M) of A after 38.46 seconds?